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Lactose is the main sugar in milk and is hydrolysed by the enzyme lactase. Lactase is essential to newborn mammals as milk is their only source of food. Most mammals stop producing lactase when they start feeding on other food sources. Humans are an exception to this because some continue to produce lactase as adults. The ability to continue producing lactase is known as lactase persistence (LP) and is controlled by a dominant allele. A number of hypotheses based on different selection pressures have been put forward to explain LP in humans.

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One hypothesis for LP in humans suggests that the selective pressure was related to some human populations farming cattle as a source of milk.

Describe how farming cattle as a source of milk could have led to an increase in LP.

[4 marks]

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Use the information provided to explain why the number of people showing LP would **rapidly** increase once selection for this condition had been established.

[2 marks]

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Lactase persistence is caused by a mutation in DNA. This mutation does **not** occur in the gene coding for lactase.

Suggest and explain how this mutation causes LP.

[2 marks]

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Turn over for the next question

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